Exact maximum lengths of cyclic orientable sequences of order nine

Prepublication draft · Complete proofs in Lean 4.19.0 · Novelty assessment and external review pending

This work presents two exact maximum theorems, their complete Lean proofs, finite certificates, and a separate Python verification. Research, auditing, and drafting used AI from the same platform. Historical novelty and independent human review remain unresolved; the prior work and scope of the contribution are discussed below.

1. Definitions and results

Let A be an alphabet of size k. Write Rw for the reversal of a word w. A word is palindromic if Rw=w and uniform if all its letters are equal. A cyclic orientable sequence of order nine, $OS_k(9)$, is a circular word s of length m such that its m length-nine windows, indexed modulo m, are distinct, none is palindromic, and no two are reversals of one another. Let $M_k(9)$ denote the largest possible length. This definition explicitly includes windows crossing the cyclic boundary.

Theorem 1. For every odd integer k≥3,

$$M_k(9) = \frac{k^9 - 4k^5 - 4k^4 + 2k^3 + 11k^2 - 6k}{2}.$$

Theorem 2. For every even integer k≥4,

$$M_k(9) = \frac{k^9 - 4k^5 + 2k^3 + k^2 - 8k}{2}.$$

No result for k=2 is asserted here. The theorems concern the maximum over all orientable sequences on the alphabet, without restricting their windows to necklaces, bracelets, supports invariant under alphabet renaming, or a family generated by an algorithm.

The proof consists of universal upper bounds (§§2–4), constructions attaining them (§§7–8), a connectivity lemma (§5), and a circuit-to-word lemma that includes the cyclic boundary (§6). The finite certificates are explicit parts of the proof. Their interpretation and the identities checked are stated; the argument does not rely on a solver's claim of optimality.

2. Reducing any orientable sequence to an omission set

Let F be the digraph whose vertices are the words in A^8 and whose arcs are the nonpalindromic words in A^9. An arc w goes from its length-eight prefix to its length-eight suffix. Reversal sends an arc u→v to the arc Rv→Ru. There are k^9-k^5 arcs, since a length-nine palindrome is determined by its first five letters.

For $v \in A^8$, define α(v)=1 if its length-seven suffix is palindromic and 0 otherwise, and define β(v) analogously for its prefix. Exactly one outgoing extension is palindromic when α=1, and exactly one incoming extension is palindromic when β=1. Thus

$$d_F^+(v) = k - \alpha(v), \quad d_F^-(v) = k - \beta(v),$$

$$\delta(v) := d_F^+(v) - d_F^-(v) = \beta(v) - \alpha(v).$$

Call vertices with δ=1 sources and those with δ=−1 sinks. Reversal exchanges these two sets.

Given an orientable sequence of length m, let J be its set of windows. Since the word is circular, J is balanced: $d_J^+ = d_J^-$ at every vertex. The set RJ is also balanced. Orientability implies J∩RJ=∅. Hence

$$H = J \cup RJ, \qquad D = F \setminus H, \qquad |H| = 2m,$$

$$RD = D, \quad \operatorname{div} D = \delta. \tag{1}$$

At a palindromic vertex p,

$$d_H^+(p) = d_J^+(p) + d_J^-(p) = 2d_J^+(p). \tag{2}$$

If p is palindromic and nonuniform, then α(p)=β(p)=0. Indeed, if both the reflection of its eight positions and that of a length-seven prefix or suffix preserved the letters, all letters would be equal. At a uniform vertex, α=β=1. Consequently, by (2), D contains at least one outgoing and one incoming arc at every nonuniform palindrome when k is odd, and at every uniform vertex when k is even. This remains true if H leaves that vertex inactive. Let T_k denote the set of palindromic vertices subject to this minimum loss.

3. A bound from potentials and charges

Lemma 3. Suppose $P:A^8 \to \mathbb{Z}$ and $\rho:A^8 \to \mathbb{Z}$ satisfy:

1. P(Rv)=−P(v);
2. ρ(v)≥0, with ρ(v)=0 outside T_k;
3. for every arc u→v of F,

$$P(u) - P(v) + \rho(u) + \rho(v) \le 2. \tag{3}$$

Then every $OS_k(9)$ satisfies

$$2m \le k^9 - k^5 - C(k),$$

$$C(k) = \sum_{\delta(v)=1} P(v) + \sum_v \rho(v). \tag{4}$$

Proof. Apply (3) to the set D in (1), rather than to a restricted family of omissions. Summing and collecting contributions at each vertex gives

$$2|D| \geq \sum_v P(v)\delta(v)$$

$$+\sum_v \rho(v)(d_D^+(v) + d_D^-(v)).$$

The first sum is twice the sum over sources, because R exchanges sources and sinks and P changes sign. The second is at least 2Σρ by the minimum loss proved in §2. The potential P need not be nonnegative. Dividing by two and using $|D|=k^9-k^5-2m$ gives (4). ∎

4. Finite certificates covering all alphabets

4.1 Equality patterns

The pattern π(w) of a word replaces its first distinct letter by 0, its next previously unseen letter by 1, and so on. It is a restricted-growth string: it begins with 0, and each subsequent symbol lies between 0 and one more than the previous maximum. Its rank r is the number of distinct symbols.

These length-n patterns are in bijection with partitions of the n positions: the blocks of a partition are ordered by first occurrence. The generator in `verify.py` extends each prefix by every allowed value, producing every partition exactly once. It enumerates the $B_8=4{,}140$ vertex patterns and $B_9=21{,}147$ arc patterns, including ranks up to nine. Their rank inventories are checked against the recurrence $S(n,r)=rS(n-1,r)+S(n-1,r-1)$, with $S(0,0)=1$. There are 52 palindromic and 21,095 nonpalindromic length-nine patterns.

Each rank-r pattern has exactly

$$(k)_r = k(k-1)\cdots(k-r+1)$$

realizations over A: one for each injection of its r ordered classes into A. There is no division by r!. If r>k, the product is zero and there are no realizations. This allows identities to be checked for all k through patterns and exact counting, without interpolation from small alphabets.

4.2 The two upper-bound certificates

The files upper_odd.json and upper_even.json contain tables for P and ρ indexed by the 4,140 patterns. Define $P(v)=P(\pi(v))$ and $\rho(v)=\rho(\pi(v))$. The verifier checks integer values, exact key coverage, antisymmetry with separate normalization of Rv, the sign and support of ρ, and (3) for all 21,095 nonpalindromic arcs. The charge support consists of nonuniform palindromes in the first file and uniform vertices in the second.

The coefficients in (4), obtained by summing P over source patterns and ρ over all patterns, are listed below. The omitted columns for ranks zero and six through nine are zero and are also checked.

Certificate	$(k)_1$	$(k)_2$	$(k)_3$	$(k)_4$	$(k)_5$
Odd	0	56	97	34	3
Even	8	38	73	30	3

Exact expansion gives

$$C_o(k) = 3k^5 + 4k^4 - 2k^3 - 11k^2 + 6k,$$

$$C_e(k) = 3k^5 - 2k^3 - k^2 + 8k. \tag{5}$$

Substitution into (4) proves the upper bounds in Theorems 1 and 2. Every literal word realizes a checked pattern, so the bounds also apply to large and composite alphabet sizes and to sequences without alphabet-renaming symmetry. The symmetry of the certificate does not restrict the object to which it applies.

5. A connectivity kernel for k≥5

Let K be the subdigraph whose arcs are length-nine words with distinct first and last letters and at least three distinct symbols. Their endpoints are nonuniform vertices. Moreover, $K \subset F$ and $RK = K$.

Lemma 4. For $k \geq 5$, K is strongly connected on all nonuniform vertices of A^8.

Proof. Call a length-eight block proper if each consecutive triple contains three distinct letters. From a nonuniform vertex v, append eight letters to reach a proper block as follows. At each step, if the current vertex has exactly two symbols, append a letter distinct from both; if it has at least three symbols, avoid its first letter and its last two letters. Each arc used has distinct first and last letters and at least three symbols, and the new vertex remains nonuniform. Each appended letter differs from the preceding two, so the final block is proper.

Given proper blocks $p = p_1 \ldots p_8$ and $q = q_1 \ldots q_8$, choose w_i successively, avoiding p_i, q_i, and the preceding two letters of w whenever they exist. At most four letters are forbidden, and $k \geq 5$. The block w is proper. When shifting p and appending w, the first and last letters of each arc are p_i and w_i; when shifting w and appending q, they are w_i and q_i. They are distinct. Every length-nine window in either concatenation contains a consecutive fragment of length at least five from one of the proper blocks, and therefore contains at least three symbols. Thus p reaches w and w reaches q in K.

Finally, to reach any nonuniform v, apply the first part to Rv and reverse that path. Since $RK = K$, this gives a path from a proper block to v. Combining the three parts, every nonuniform vertex reaches every other one. ∎

6. From a balanced support to one orientable word

Lemma 5 (a literal circuit compatible with reversal). Let H be a nonempty subset of the arcs of F, closed under reversal, balanced at every vertex, and with $d_H^+(p)$ even at every palindromic vertex. If H is strongly connected on its active vertices, there is a cyclic orientable sequence with exactly |H|/2 windows.

Proof. Consider directed trails that do not repeat arcs or use both members of a reversal pair. For a trail from a to b, its doubled set D, consisting of its arcs and their reversals, has divergence at v equal to

$$1_{v=a} - 1_{v=b} - 1_{v=Ra} + 1_{v=Rb}.$$

If b≠a and b is not palindromic, this divergence at b is −1 or −2. Since H is balanced, H\D retains an outgoing arc at b. If b is palindromic, $d_D^+(b)$ is odd: it is the sum of the trail's incoming and outgoing incidences at b, and incoming incidences exceed outgoing ones by one. Since $d_H^+(b)$ is even, an outgoing arc again remains. In either case the trail can be extended by an arc whose reversal pair has not yet been used. By finiteness, a longest such trail containing a given initial arc must end at a. This yields a nonempty literal circuit.

Choose a circuit C of maximum length among such circuits. Its doubled set D is balanced and has even out-degrees at palindromic vertices. The residual H\D retains these properties and reversal closure. If a residual arc leaves a vertex visited by D, rotate C or reverse its entire traversal so that it begins at that vertex. The preceding argument yields a nonempty residual circuit there. Splicing the circuits contradicts maximality of C.

Therefore, every outgoing arc of a vertex visited by D belongs to D. Balance of D ensures that the head of such an arc also has an outgoing arc in D and is visited. Strong connectivity propagates this property to every active vertex of H, so every arc of H belongs to D. The circuit C contains exactly one member of each reversal pair.

The length-eight overlaps, including the last-to-first overlap, reconstruct a circular word whose windows are exactly the arcs of C. They are distinct, nonpalindromic, and contain no reversal pair. Its length is |H|/2. ∎

The lemma uses literal strong connectivity, which is proved for the concrete residuals below. Reversal-compatible circuit techniques have antecedents, including the alternating incidences of Burris–Motta–Shipman [BMS, §3]; no novelty is claimed for this general tool.

7. Construction for odd alphabets

The file omissions_odd.json lists a set O of 190 length-nine patterns. Let D_o be the set of all their realizations over A. The finite verification certifies the following properties.

- O contains no palindrome and is closed under reversal followed by normalization.
- Its rank counts are 56, 97, 34, and 3 for ranks 2, 3, 4, and 5, respectively. Hence $|D_o|=C_o(k)$.
- For every length-eight pattern v of rank r, setting χ(w)=1 if π(w)∈O and zero otherwise,

$$A(v) : = \sum_{x=0}^{r-1} (\chi(vx) - \chi(xv)) = \delta(v),$$

$$B(v) : = \chi(vr) - \chi(rv) = 0. \tag{6}$$

- At every nonuniform palindrome, $\Sigma_{x<r}\chi(vx)=1$ and $\chi(vr)=0$. At a uniform vertex, no outgoing extension is omitted, whether its appended symbol is repeated or new.
- Every omitted pattern of rank at least three has equal first and last letters.

These properties are checked on all vertex patterns, not just those appearing in a search for a particular k. For a literal realization of v, each of the k−r unused letters produces the same pattern as the new symbol r, while the r existing letters produce the terms of A. Thus

$$\operatorname{div} D_o(v) = A(v) + (k-r)B(v) = \delta(v).$$

Consequently, $H_o=F\backslash D_o$ is balanced and closed under reversal. If k is odd, its out-degree at every palindromic vertex is k−1: at a nonuniform palindrome, exactly one of the k outgoing arcs of F is lost; at a uniform vertex, none of the k−1 outgoing arcs of F is lost. This degree is even.

The last property of O guarantees $K\subset H_o$. For k≥5, Lemma 4 connects all nonuniform vertices. Uniform vertices attach through the arcs a^8b and ba^8 with b≠a, which are not omitted. Thus H_o is strongly connected on its active vertices. Lemma 5 gives an orientable sequence of length $(k^9-k^5-C_o(k))/2$, attaining the upper bound.

For k=3, the five-letter lemma is not invoked. The formal proof enumerates all 6,561 vertices of H_o and verifies a certificate of decreasing distances toward 0^8. Reversing a path from Rv to 0^8 gives a path from 0^8 to v, so H_o is strongly connected. It has 18,522 arcs, and Lemma 5 gives length 9,261. As additional, separately checked evidence, word3.json contains a literal word of length 9,261. Its symbols and all 9,261 cyclic windows are checked, including the eight crossing the cyclic boundary, with no palindrome, repetition, or reversal pair. The upper bound already proved equals 9,261 at k=3. This completes the proof of Theorem 1. ∎

8. Construction for even alphabets

8.1 Paths from sources to sinks

Every source has the form s(a,b,c,d,e)=abcdcbae, with a,b,c,d,e∈A. Its length-seven prefix is palindromic. Its length-seven suffix is palindromic exactly in the exceptional case E: c=a, d=b, e=b. Excluding E leaves k^5-k^2 sources, with an injective parametrization.

Call a source short if it satisfies L: c=a and e=d. Excluding E then requires b≠d. There are k^3-k^2 short sources. For each, omit the arc

$$W(a,b,d) = abadabada \quad (b \neq d).$$

For each of the remaining k^5-k^3 sources, satisfying neither E nor L, omit the three consecutive arcs

$$A_1 = abcdcbaea, \quad A_2 = bcdcbaeab, \quad A_3 = cdcbaeabc. \tag{7}$$

Let D_0 be the union of these omissions. We show that the arcs counted are distinct even when several parameters coincide. In the long families, the indicators for a palindromic length-seven prefix and suffix are respectively (1,0), (0,0), and (0,1); in W they are (1,1). These distinguish the four families. Using zero-based indices, the inverses for A_1, A_2, and A_3 recover the parameters as

$$(w_0, w_1, w_2, w_3, w_7),$$

$$(w_5, w_0, w_1, w_2, w_6),$$

$$(w_4, w_3, w_0, w_1, w_5),$$

while W recovers (a,b,d)=(w_0,w_1,w_3). Each family is therefore injective. The arcs A_1 and A_3 are palindromic exactly under E, A_2 under L, and W when b=d; all these cases are excluded.

The map f(a,b,c,d,e)=(c,b,a,e,d) is an involution preserving the long-source parameter set. It sends RA_1 to A_3, RA_2 to A_2, and RA_3 to A_1 at the transformed parameters. For W, reversal exchanges b and d. Therefore $RD_0=D_0$.

The long path ends at dcbaeabc. Its reversal is the source s(f(a,b,c,d,e)). The short path ends at badabada, whose reversal is the short source with b and d exchanged. Every source and every sink is used exactly once. Interior incidences cancel even if different paths share vertices, because their arcs are distinct. Hence

$$\operatorname{div} D_0 = \delta,$$

$$|D_0| = 3(k^5 - k^3) + (k^3 - k^2) = 3k^5 - 2k^3 - k^2. \tag{8}$$

No arc of D_0 is incident to a palindromic vertex. The source and sink endpoints are not palindromic because $\delta \neq 0$. Requiring an interior vertex of (7) to be palindromic forces all its letters, and hence all parameters, to coincide, an excluded case. Every arc of D_0 has equal first and last letters.

As an additional exhaustive check, `verify.py` verifies the identities on all 52 equality patterns of five parameters and checks on all 21,147 length-nine patterns that D_0 is equivalent to the predicate

$$w \neq Rw, \quad w_0 = w_8,$$

$$[\mathrm{pal}(w_0 \cdots w_6) \ \vee$$

$$\mathrm{pal}(w_2 \cdots w_8) \ \vee$$

$$(w_0 = w_4,\ w_1 = w_3,\ w_5 = w_7)].$$

It recovers 144 patterns, with counts 38, 73, 30, and 3 in ranks 2, 3, 4, and 5. It also checks (6) for this predicate, giving a second check of the divergence in every degenerate case.

8.2 Repairing parity at uniform vertices

Since k is even, pair the alphabet letters arbitrarily. For each pair {a,b}, also omit the 16 length-nine windows of the circular word a^8b^8. They are

$$a^i b^{9-i}, \quad b^i a^{9-i}, \qquad 1 \le i \le 8.$$

These arcs are distinct, nonpalindromic, closed under reversal, and balanced because they come from a circular word. They have distinct first and last letters, so they are disjoint from D_0. Different pairs use disjoint alphabets. Each uniform vertex loses exactly one outgoing and one incoming arc; no other palindromic vertex is incident to these arcs. Write D_m for their union. Then $|D_m|=16(k/2)=8k$.

The residual $H_e=F\setminus(D_0\cup D_m)$ is balanced and closed under reversal. Its out-degree is $k-2$ at uniform vertices and k at nonuniform palindromes, both even. Its size is

$$|H_e| = k^9 - k^5 - (3k^5 - 2k^3 - k^2 + 8k).$$

8.3 Connectivity and the quaternary boundary case

Every arc of K survives: D_0 has equal first and last letters, while D_m uses only two symbols. For $k\ge6$, Lemma 4 connects the nonuniform vertices. A uniform vertex a^8 attaches through a^8c and ca^8, choosing c distinct from a and its paired letter; these arcs lie in neither D_0 nor D_m. Lemma 5 produces an orientable sequence of length $|H_e|/2$, attaining the bound in Theorem 2.

For k=4, the formal proof verifies a decreasing-distance certificate for all 65,536 vertices of H_e. The same reversal argument proves strong connectivity; its 258,160 arcs and Lemma 5 give length 129,080. As additional evidence, word4.json contains a word of length 129,080. The verifier checks its symbols and all 129,080 cyclic windows, including the eight crossing the boundary. It finds no palindrome, repetition, or reversal pair. The bound in §4 is exactly 129,080. This additional word check does not require trust in its generating algorithm or identification of its support with H_e. This completes Theorem 2. ∎

9. Scope of the contribution and quantitative comparisons

The candidate contribution is the exact value of $M_k(9)$ in both domains, through two bound certificates and constructions attaining them. No novelty is claimed for overlap graphs, reversal, palindromic parity, source-to-sink paths, Lempel lifting, necklaces, or alternating circuits. The located literature provides antecedents for these tools [MW-M, MW-A, GS, BMS].

Two Mitchell–Wild papers must be distinguished. [MW-M], published in March 2026, has preprint identifier 2507.02526 and gives the following bounds in Theorem 2.13(iv):

$$U_o = (k^9 - 3k^5 - 2k^4 + k^3 + 3k^2)/2,$$

$$U_e = (k^9 - 3k^5 + k^3 + k^2 - 2k)/2.$$

[MW-A], arXiv:2603.18646v2 from August 2026, is a different work. To compare with its complete support construction, including short-period orbits, our count of its excluded length-eight words gives

$$T_o = 4k^4 + 4k^3 - 2k^2 - 3k - 2,$$

$$T_e = 4k^4 + 16k^3 - 2k^2 - 9k - 6,$$

$$B = (k^9 - kT)/2.$$

Here B is a specialization computed in this work from a published construction; it is not attributed to those authors as an exact maximum theorem. The conditions and count are explained in the appendix. The construction in [GS] gives $L=(k^9-9k^5+6k^2+2k)/2$. The reproduced comparison is:

k	GS construction: L	Full MW-A support: B	Proved maximum	MW-M bound: U
3	8,778	9,237	9,261	9,423
4	126,516	127,124	129,080	129,572
5	962,580	969,230	969,310	971,350
6	5,003,970	5,013,324	5,023,506	5,027,304

The improvement over B is $4k(k-1)$ for odd k and $(16k^4-8k^2-14k)/2$ for even k. The improvement of the upper bound U is $k(k-1)^2(k^2+4k+6)/2$ for odd k and $(k^5-k^3+6k)/2$ for even k. These are positive in the stated domains.

The literature search did not locate these equalities in the sources actually examined. This is a limited conclusion, separate from establishing priority. The 1993 chapter has been read in full: it treats the binary case and does not contain the formulas presented here. Review of the public *Concatenation trees* manuscript was extended to its general theorems; equivalence with the 2026 publisher version remains unconfirmed. BIBLIOGRAPHY.md records versions, passages, resolved leads, and remaining gaps.

10. Proof dependencies and reproduction

The two upper-bound tables, the odd selection, the even base, and the boundary connectivity certificates are transcribed as explicit Lean data. Coverage of the finite enumerations, local properties, their sums, the universal upper bound, constructions, connectivity, and cyclic reconstruction form a complete formal chain. `FullTargets.lean` proves both maximum statements throughout their respective domains.

`python3 reproduce_all.py --lean /path/to/lean-4.19.0/bin/lean` checks the five original data files, checks the additional transcriptions, recompiles every source file from scratch, and audits its axioms. Only the standard axioms `propext`, `Quot.sound`, and `Classical.choice` are allowed. Instructions and records are linked from README.md; FORMALIZATION.md describes the definitions and their correspondence.

`python3 reproduce.py` performs only the separate exact checks of tables, constructions, words, and prior-work comparisons. It is not by itself a universal proof. The Lean proof does not treat Python as an oracle. Hashes identify the checked objects. The remaining trust assumptions of the formalization concern the Lean kernel and standard library, and the fidelity of the model; these should be examined during external review.

11. Provenance, AI use, and review status

Research and candidate generation, auditing, formalization, code, and drafting were performed using AI from the same platform in separate tasks. The verifier's independence is at the implementation level; it does not constitute external human or institutional review. The original mathematical data are preserved, and their provenance and transcription are documented in DATA_PROVENANCE.json and FORMALIZATION.md.

No human authorship or supervision is attributed where none has occurred. Before journal submission, actual human responsibility for reviewing the proof, sources, and limitations must be established, and AI use must be disclosed at this scope. This version is prepared for external review and includes the complete formalization of Theorems 1 and 2. Bibliographic and publication status are maintained in STATUS.md, without claiming worldwide priority.

Appendix: exact counts for earlier supports

The construction in [GS, §6] selects an orientation from each pair of asymmetric rotation classes. A length-nine word is excluded if it is fixed by some reflection $i \mapsto t-i \pmod 9$. The union of these nine fixed-point sets contains $9k^5-6k^2-2k$ words; dividing the size of the complement by two gives L.

In [MW-A, Theorem 3.1 and Lemmas 5.3–5.5], the negative-orientable support at length eight selects half by pseudoweight and one orientation from each remaining pair in the central layer. Rotation classes symmetric under reversal with negation lie in that central layer and are excluded. Their union of words is described by the eight reflections $x_i=-x_{t-i}$. Difference lifting has k preimages per word, giving $B=k(k^8-T)/2$.

To reproduce T and the ordinary count, apply inclusion–exclusion over every nonempty subset of reflections. Each intersection is a system of signed equalities. In each connected component, a free letter has k choices unless a cycle of signs forces $x=-x$: in $\mathbb{Z}/k\mathbb{Z}$ this gives $\gcd(2,k)$ choices. If an intersection has f free components and t components with this constraint, its contribution is $k^f \gcd(2,k)^t$, with the usual inclusion–exclusion sign. Without negation, each component contributes k. The algorithm enumerates 255 and 511 subsets, respectively, and expands the preceding polynomials exactly. It counts words, rather than multiplying the number of necklaces by a presumed constant length, so it retains short periods and applies to composite k.

References

- **[MW-M]** Chris J. Mitchell and Peter R. Wild, *New orientable sequences*, Acta Informatica 63, article 14 (18 March 2026). Publisher version; corresponding preprint 2507.02526. Theorem 2.13(iv), Lemma 2.10(iv).
- **[MW-A]** Chris J. Mitchell and Peter R. Wild, *Constructing orientable and negative orientable sequences with asymptotically optimal period*, arXiv:2603.18646v2, 14 August 2026. Theorem 3.1, Lemmas 5.3–5.5, Theorems 5.6 and 5.8. This is distinct from [MW-M].
- **[GS]** Daniel Gabrić and Joe Sawada, *Constructing k-ary Orientable Sequences with Asymptotically Optimal Length*, Designs, Codes and Cryptography 93 (2025), 2349–2367. Examined preprint, §6, Theorems 11–14.
- **[BMS]** Christie S. Burris, Francis C. Motta, and Patrick D. Shipman, *An Unoriented Variation on de Bruijn Sequences*. Examined version, §3 for alternating incidences; §5 concerns coverage with repetitions, which differs from the maximum defined here.
- **[DMRW]** Z.-D. Dai, K. M. Martin, M. J. B. Robshaw, and P. R. Wild, *Orientable Sequences*, in M. J. Ganley (ed.), Cryptography and Coding III (1993), pp. 97–115. Consulted volume.
- **[SSTW]** Joe Sawada, Jackson Sears, Andrew Trautrim, and Aaron Williams, *Concatenation trees: A framework for efficient universal cycle and de Bruijn sequence constructions*. Consulted manuscript; DOI of the publisher version awaiting comparison.

Further references, data provenance, and review status are recorded in BIBLIOGRAPHY.md, DATA_PROVENANCE.json, and STATUS.md.

Project credit

Project initiation and coordination: **@AIOvernight**.

A night full of tokens + RTX 5090